TWO-DAY HANDS-ON WORKSHOP / B.TECH CLOUD COMPUTING LAB

CLOUD COMPUTING

Learn the foundations. Connect the systems.
Build with confidence.

DAY 01

FOUNDATIONS + NETWORKING

CONCEPTS → CONNECTIONS

FIRST HALF / SESSION 01

Cloud Computing + Setup

Get the environment ready and understand the cloud landscape.

- Cloud computing: what it is and why it matters
- IaaS, PaaS and SaaS
- Public, private and hybrid cloud
- Virtual machines vs. containers
- Set up VirtualBox / VMware and Ubuntu
- Linux terminal basics
- Quick tour: AWS, Azure and Google Cloud

SECOND HALF / SESSION 02

Computer Networks

Understand how a user, a network and a cloud server communicate.

- IP addresses: public vs. private
- DNS and HTTP / HTTPS
- TCP / UDP and ports
- Routers, gateways and NAT
- Firewalls and cloud security groups
- Practice: ping and ipconfig / ip addr
- Practice: nslookup, curl and route tracing

DAY 01 / TAKEAWAY

Students understand cloud basics, have a working lab environment, and can explain how a browser reaches a server.

PREP BEFORE THE WORKSHOP Download the Ubuntu image, install VM software, check laptop storage / RAM, and keep a backup environment ready.

DAY 02

PRACTICALS + MINI PROJECT

DAY 02 **BUILD • DEPLOY • SECURE • PRESENT** **HANDS-ON DAY**

FIRST HALF / SESSION 03

Cloud Lab Practicals

Work through core experiments from the syllabus.

- Create / launch a Linux virtual machine
- Explore OS settings and VM resources
- Install a C compiler and run a simple program
- Build and run a basic web application
- Demonstrate Google App Engine / PaaS
- Introduce AWS EC2, storage and access rules
- Quick demos: serverless, Hadoop, databases and chatbots

SECOND HALF / SESSION 04

Cloud Deployment Challenge

Team up and turn the concepts into a working demo.

- Form teams of 3–4 students
- Build a portfolio or campus webpage
- Run it on a VM or approved cloud environment
- Test access from another permitted device
- Secure it: restrict ports and protect credentials
- Check logs and troubleshoot one issue
- Present a 3-minute live demo per team

01 BUILD → **02 DEPLOY** → **03 SECURE** → **04 DEMO**

DAY 02 / TAKEAWAY

Every team demonstrates a working application, explains its network path, and identifies basic security controls.

KEEP IT SMOOTH

Prepare VM images and sample code in advance. If cloud accounts or internet are unavailable, use a local VM. Stop or delete temporary cloud resources after the workshop.